

DRISHTI SAT-SA — Architecture (PS 26157)

Team CyberNova · SIH 2026 · Offline explainable supervisory analytics for NCIIPC. SHA3-256 live. ML-DSA-65 next. Engine never closes a ticket.

1. Purpose and boundary

NCIIPC already receives SOC output. SAT-SA judges whether that output is work. DRISHTI sits after Splunk / QRadar / Sentinel. It does not monitor live, collect telemetry, or close tickets. Examiners Accept / Reject / FP every queued finding. Offline: no internet, no cloud, no SaaS, no hosted models. USB ZIP / JSON / CSV in → dashboard + sealed JSON/CSV/SQLite out. Out of scope: CSE SOC, real-time, SIEM, multi-tenant SOC, national sensor grid, continuous logs.

2. Six layers

CSE extract (USB) → pandas store + SHA3 → dual engine (12 EG + 8 NS + Isolation Forest + TF-IDF) → SPS 0–100 + sector median + Z>2 → Top-100 queue → Trinity + SHA3 chain → Supervisor UI

Layer	What happens
Ingest	Schema check, provenance hash, no raw logs / PII
Store	Alerts, cases, assets, analysts, playbooks, compliance
Detect	Rules first (auditable); ML only as an extra flag
Score	Capability concern rates → SPS; peer by sector
Prioritise	100 rows a person can finish; engine never decides
Trust	Rationale · Evidence · Confidence · SHA3-256 (ML-DSA-65 slot)

Runtime: Python 3.11, FastAPI, pandas, scikit-learn. UI = static SPA, relative /api/*, no CDN. Hardware: 4 vCPU, 16 GB RAM, 100 GB SSD, no GPU. Network: docker run --network=none.

3. Dual engine (PS §3)

A. Execution gaps — evidence contradicts the KPI.

ID	Signal	Conf
EG-01	Critical ∧ dwell<10 ∧ steps<2 ∧ ¬escalation	0.90
EG-02	Critical closed, no escalation	0.85
EG-03	TF-IDF cosine > 0.95, same analyst	0.91
EG-04	Same asset+type ≥5 times, no RCA	0.80
EG-05	Critical asset, 0 alerts, peers noisy	0.93
EG-06	Fast MTTR + templates + low escalation	0.88
EG-07	Critical/High, 0 investigation steps	0.82
EG-08	Escalation after SLA	0.75
EG-09	Critical notes <20 chars	0.70
EG-10	Malware closed without evidence	0.80
EG-11	Dwell exceeds severity SLA	0.75
EG-12	Analyst load >3× team median	0.65

B. Negative space — NS-01 silent critical asset · NS-02 missing MITRE/category vs sector · NS-03 zero critical escalations · NS-04 volume <10% peer median · NS-05 silent analyst · NS-06 missing control evidence · NS-07 no night/weekend alerts · NS-08 critical-asset coverage gap.

ML (PS §5): sklearn IsolationForest n=200, contamination=0.15 on (MTTR, escalation_rate, investigation_steps, silent_rate, fast_critical, night_rate). Fit on this extract only. TF-IDF cosine > 0.95 for templates. Rules fire first. Update = USB re-ingest.

4. SPS, queue, trust, validation

Eight NCIIPC capabilities: Detection 20%, Investigation 20%, Escalation 15%, IR 15%, SecOps 10%, Governance 10%, Discipline 5%, Resilience 5%. SPS = weighted concern rate; high = more attention. Sector trimmed median (10% tails), Z>2 peer flag. Queue score = confidence × severity_weight × (1+cap_weight) × (1+0.15|z|) with a per-rule cap.

Every finding: Rule ID, Confidence, Rationale, Evidence, capability, SHA3-256, previous chain hash, HMAC seal. Production swap: HMAC → ML-DSA-65. Examiner decisions append to the same chain.

Demo (measured): 12 CSEs · 3,378 alerts · 235 findings · Top 100. Planted ground_truth.csv recovered 38/40. Design target F1>0.85 is for a larger labelled extract — not claimed as a result of this demo.